

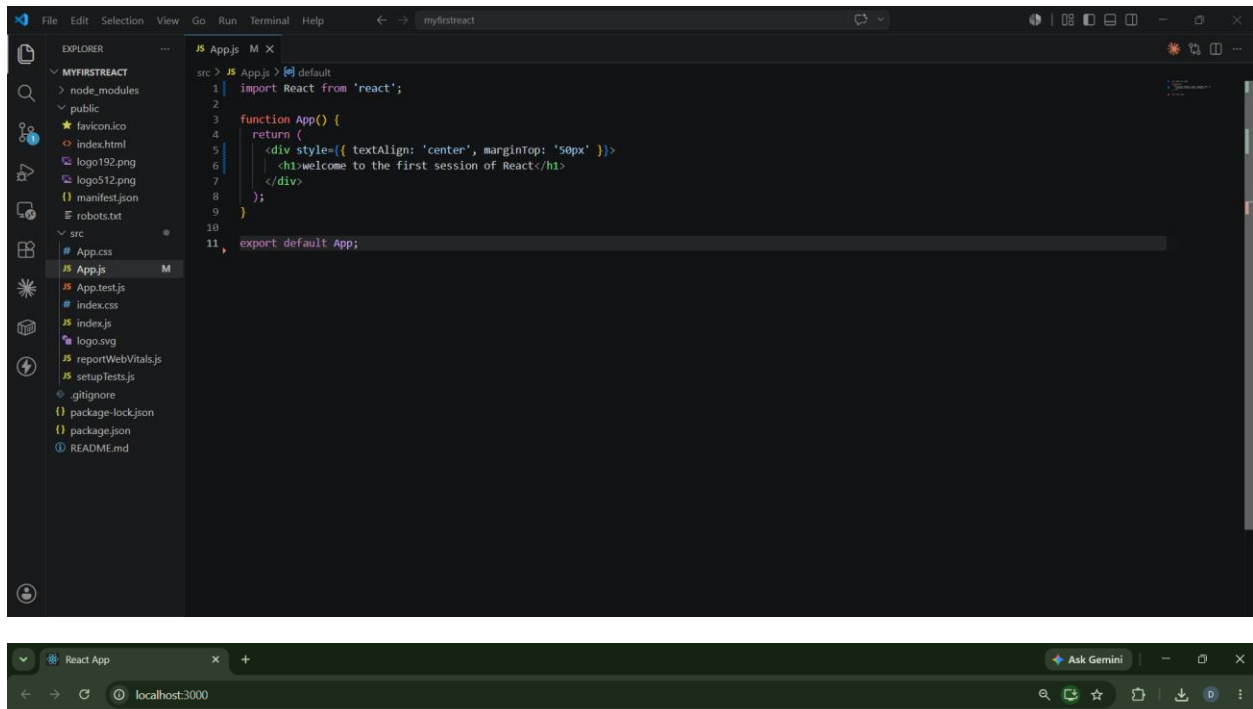
ReactJS HOL

NAME: DURGASREE AVVARU

EMAIL: 2300031591cseh@gmail.com

ReactJS HOL-1

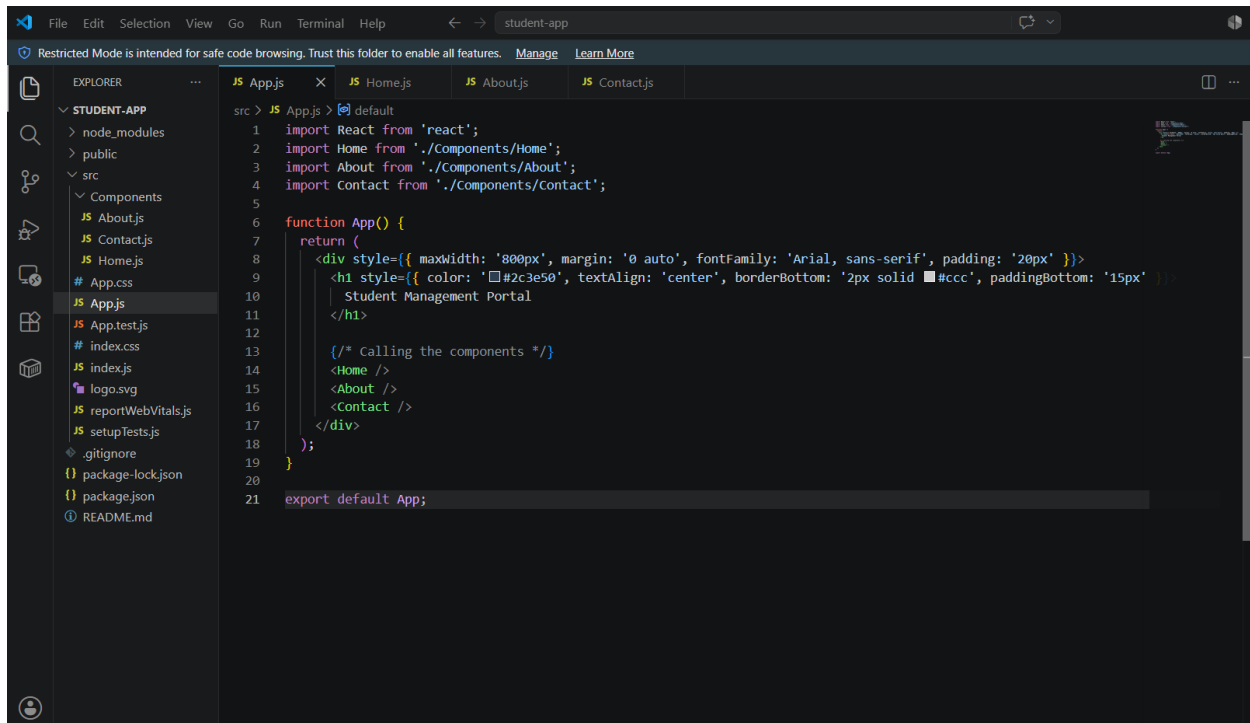
Create a new React Application with the name “myfirstreact”, Run the application to print “welcome to the first session of React” as heading of that page.



welcome to the first session of React

ReactJS HOL-2

Create a react app for Student Management Portal named StudentApp and create a component named Home which will display the Message “Welcome to the Home page of Student Management Portal”. Create another component named About and display the Message “Welcome to the About page of the Student Management Portal”. Create a third component named Contact and display the Message “Welcome to the Contact page of the Student Management Portal”. Call all the three components.



```
1 import React from 'react';
2 import Home from './Components/Home';
3 import About from './Components/About';
4 import Contact from './Components/Contact';
5
6 function App() {
7   return (
8     <div style={{ maxWidth: '800px', margin: '0 auto', fontFamily: 'Arial, sans-serif', padding: '20px' }}>
9       <h1 style={{ color: '#2c3e50', textAlign: 'center', borderBottom: '2px solid #ccc', paddingBottom: '15px' }}>
10         Student Management Portal
11       </h1>
12
13       <div>
14         <Home />
15         <About />
16         <Contact />
17       </div>
18     </div>
19   );
20 }
21 export default App;
```



Student Management Portal

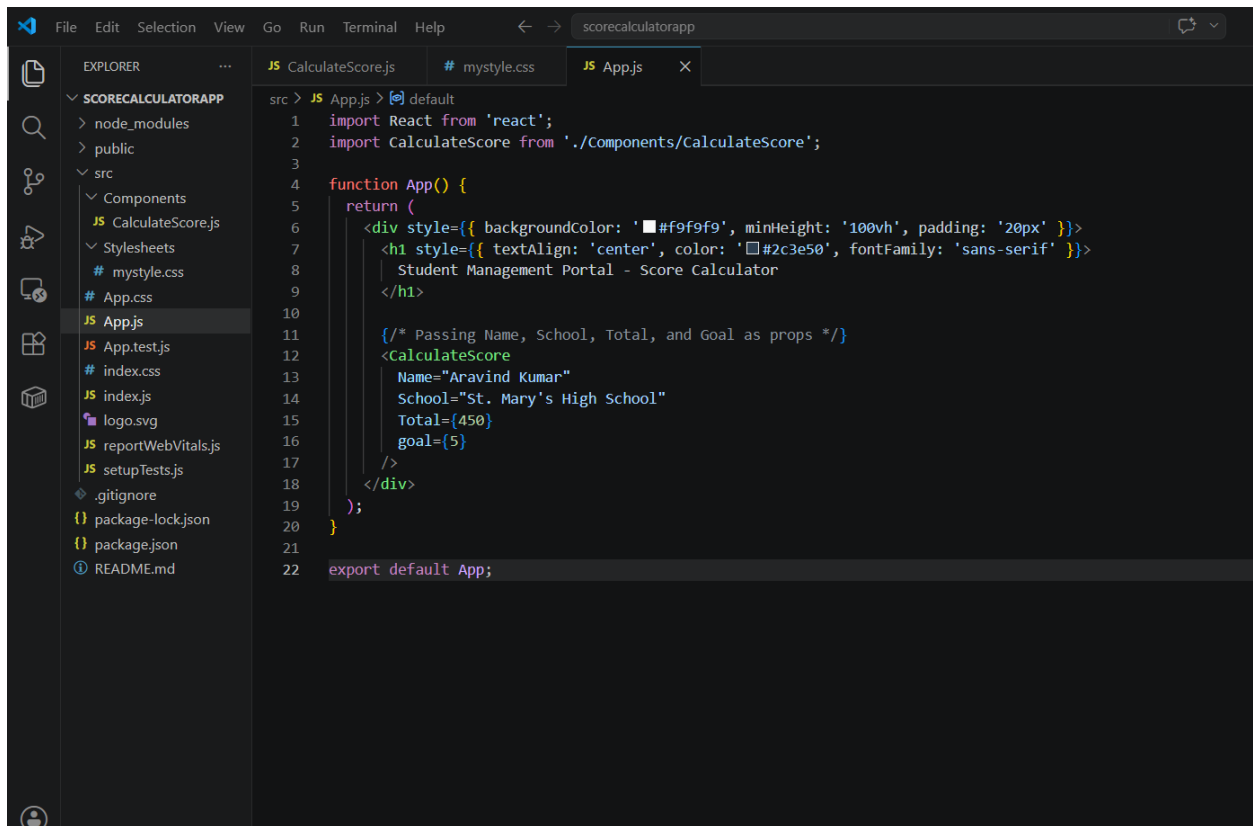
Welcome to the Home page of Student Management Portal

Welcome to the About page of the Student Management Portal

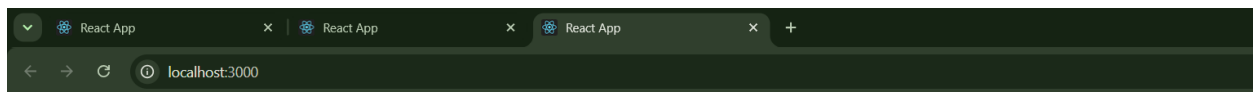
Welcome to the Contact page of the Student Management Portal

ReactJS HOL-3

Create a react app for Student Management Portal named scorecalculatorapp and create a function component named “CalculateScore” which will accept Name, School, Total and goal in order to calculate the average score of a student and display the same.



```
src > JS App.js > default
1  import React from 'react';
2  import CalculateScore from './Components/CalculateScore';
3
4  function App() {
5    return (
6      <div style={{ backgroundColor: '#f9f9f9', minHeight: '100vh', padding: '20px' }}>
7        <h1 style={{ textAlign: 'center', color: '#2c3e50', fontFamily: 'sans-serif' }}>
8          Student Management Portal - Score Calculator
9        </h1>
10
11        {/* Passing Name, School, Total, and Goal as props */}
12        <CalculateScore
13          Name="Aravind Kumar"
14          School="St. Mary's High School"
15          Total={450}
16          goal={5}
17        />
18      </div>
19    );
20  }
21
22  export default App;
```

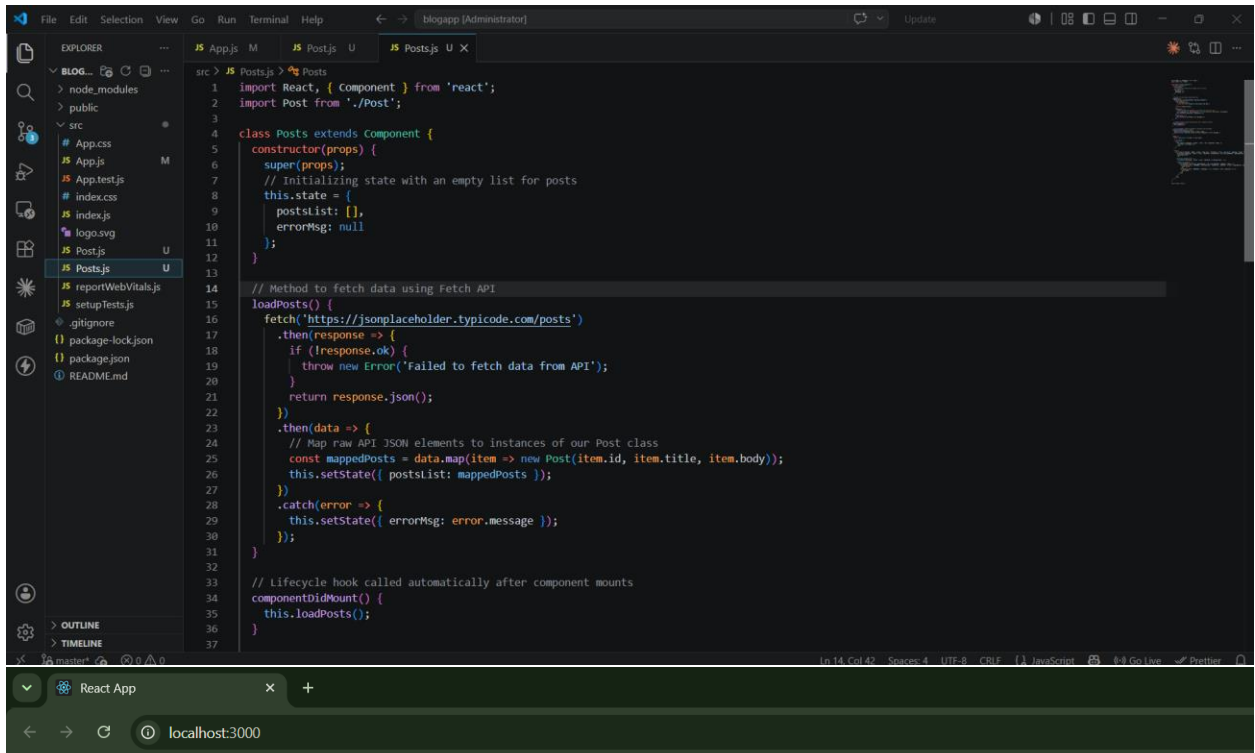


Student Management Portal - Score Calculator

Student Performance Summary	
Student Name:	Aravind Kumar
School:	St. Mary's High School
Total Marks:	450
Goal (Subjects):	5
Average Score: 90.00	

ReactJS HOL -4

1. Create a new react application using *create-react-app* tool with the name as “blogapp”



Blog Application

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Ea Molestias Quasi Exercitationem Repellat Qui Ipsa Sit Aut

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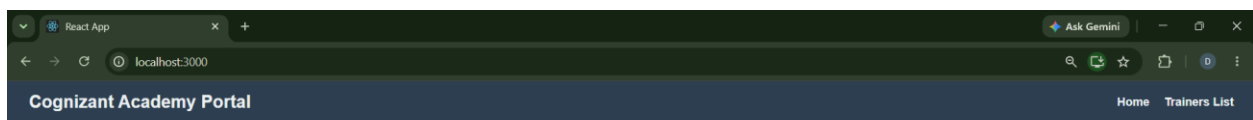
ullam et saepe reiciendis voluptatem adipisci sit amet autem assumenda provident rerum culpa quis hic commodi nesciunt rem tenetur doloremque ipsam iure quis sunt voluptatem rerum illo velit

ReactJS HOL –5

Cognizant Academy teams want to maintain a list of trainers along with their expertise in a SPA using React as the technology. You are assigned the task of creating this React app.

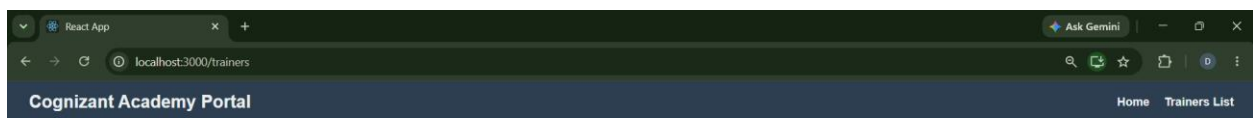
```
File Edit Selection View Go Run Terminal Help ← → trainers-app [Administrator]
EXPLORER
  TRAINERS-APP
    > node_modules
    > public
    > src
      App.css
      JS App.js
      JS AppTest.js
      JS Home.js
      JS index.css
      JS index.js
      JS logo.svg
      JS reportWebVitals.js
      JS setupTests.js
      JS Trainer.js
      JS TrainerDetails.js
      JS TrainerList.js
      JS TrainersMock.js
      .gitignore
    package-lock.json
    package.json
    README.md
  OUTLINE
  TIMELINE

src > JS App.js > default
1 // src/App.js
2 import React from 'react';
3 import { BrowserRouter, Routes, Route, Link } from 'react-router-dom';
4 import Home from './Home';
5 import TrainerList from './TrainerList';
6 import TrainerDetail from './TrainerDetails';
7 import { trainersMockData } from './TrainersMock';
8
9 function App() {
10   return (
11     <BrowserRouter>
12       <div style={{ fontFamily: 'Arial, sans-serif' }}>
13         /* Navigation Header Bar */
14         <nav style={{
15           backgroundColor: '#2c3e50',
16           padding: '15px 30px',
17           display: 'flex',
18           justifyContent: 'space-between',
19           alignItems: 'center'
20         }}>
21           <h1 style={{ color: '#ecf0f1', margin: 0, fontSize: '1.5rem' }}>Cognizant Academy Portal</h1>
22           <div>
23             <Link to="/" style={{ color: '#ecf0f1', textDecoration: 'none', marginRight: '20px', fontWeight: 'bold' }}>Home</Link>
24             <Link to="/trainers" style={{ color: '#ecf0f1', textDecoration: 'none', fontWeight: 'bold' }}>Trainers List</Link>
25           </div>
26         </nav>
27
28         /* Core Route Switching Architecture */
29         <div style={{ padding: '20px' }}>
30           <Routes>
31             <Route path="/" element={<Home />} />
32             <Route path="/trainers" element={<TrainerList trainers={trainersMockData} />} />
33             <Route path="/trainers/:id" element={<TrainerDetail trainers={trainersMockData} />} />
34           </Routes>
35         </div>
36       </div>
37     </BrowserRouter>
38   );
39 }
```



Welcome to Cognizant Academy

Managing and tracking enterprise technical training streams and expert trainers.

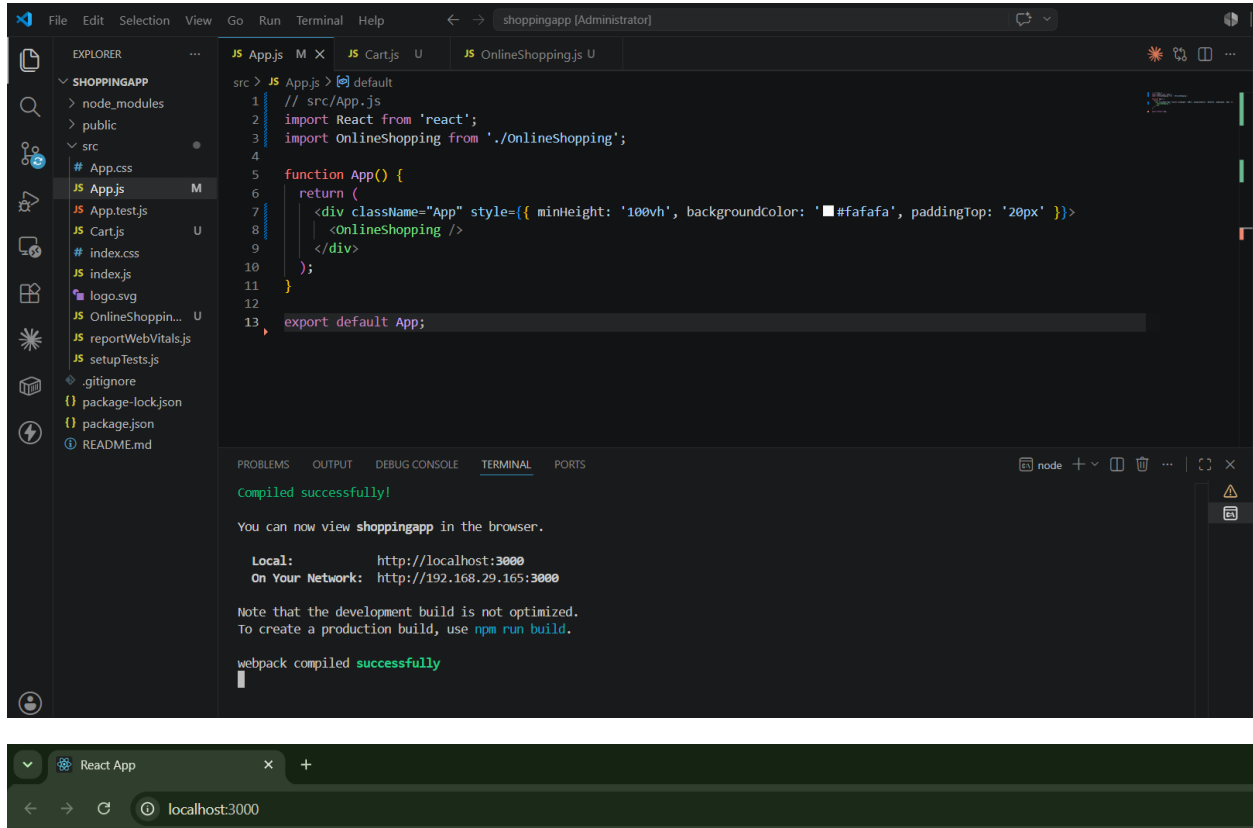


Academy Expert Trainers

- [John Doe](#) — Java Full Stack
- [Jane Smith](#) — Cloud & DevOps
- [Robert Johnson](#) — Data Science
- [Emily Davis](#) — DotNet Microservices

ReactJS HOL-6

Create a React Application named “shoppingapp” with a class component named “OnlineShopping” and “Cart”.



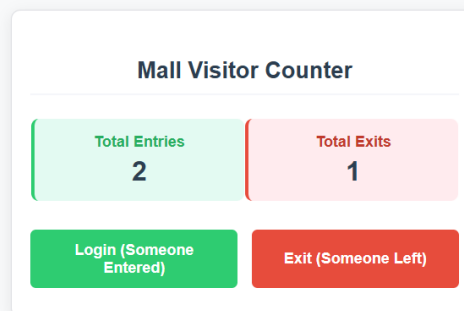
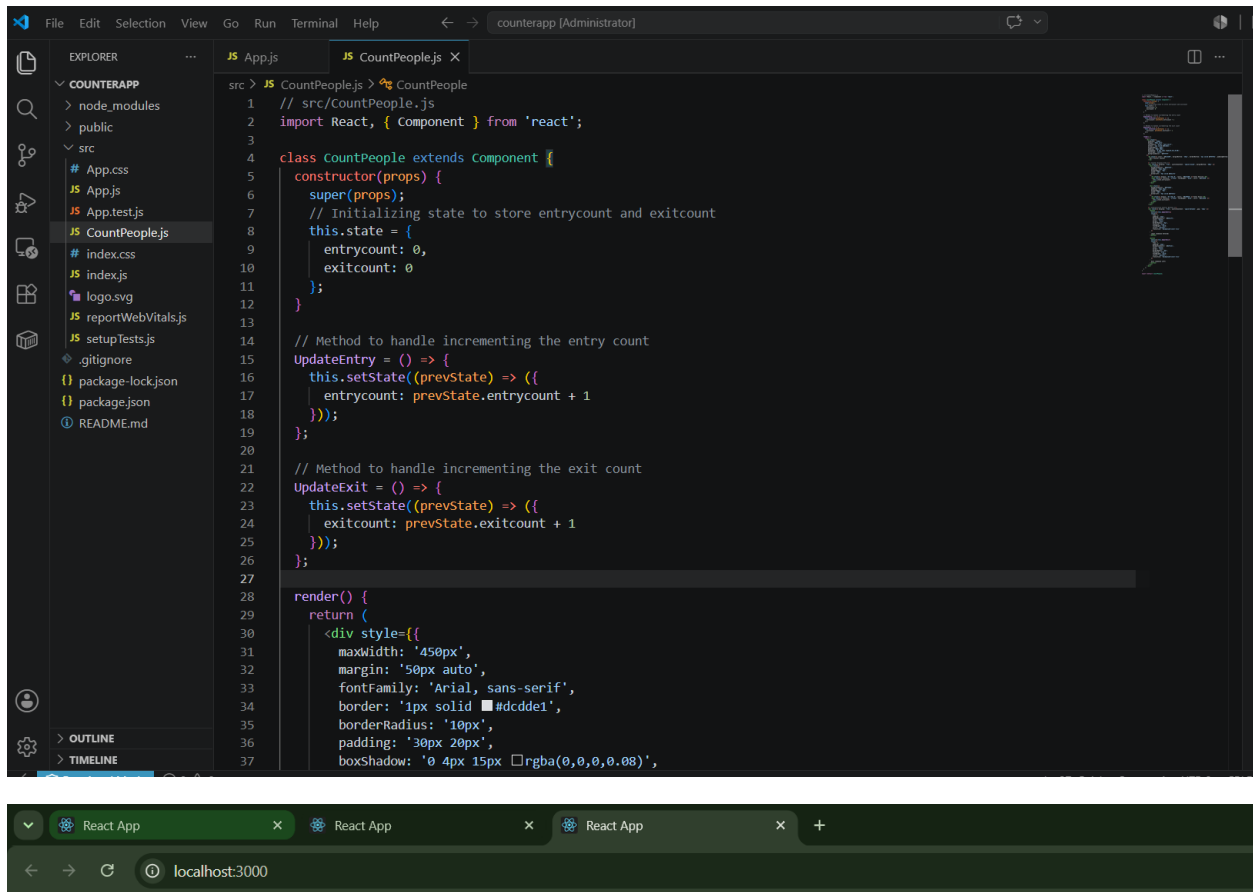
Online Shopping Cart	
Item Name	Price (INR)
Laptop	Rs. 75,000
Smart Phone	Rs. 35,000
Headphones	Rs. 3,000
Smart Watch	Rs. 8,500
Tablet	Rs. 22,000

ReactJS HOL –7

Create a React App “counterapp” which will have a component named “CountPeople” which will have 2 methods.

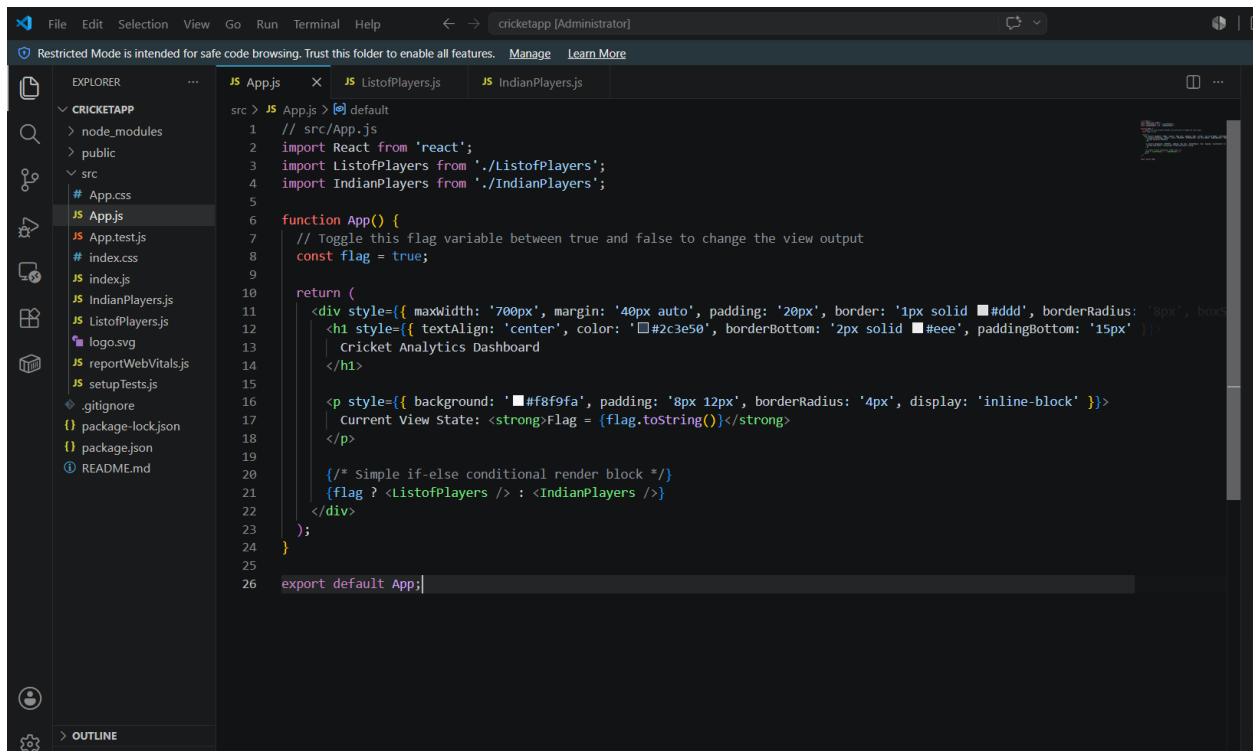
UpdateEntry() → which will display the number of people who entered the mall.

UpdateExit() → which will display the number of people who exited the mall.

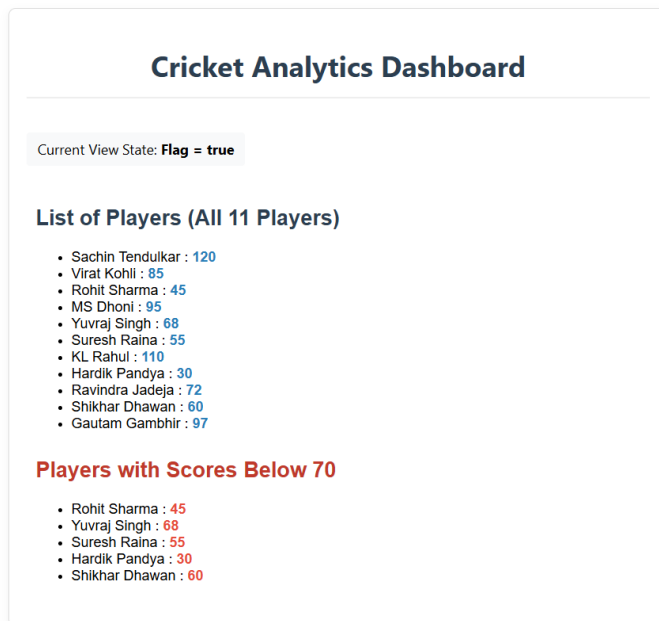
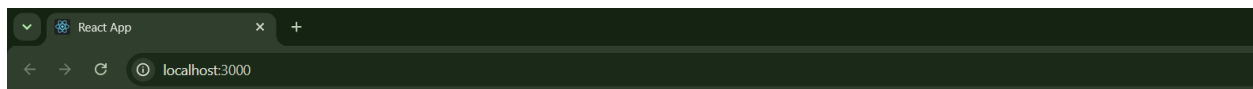


ReactJS HOL-8

Create a React Application named "cricketapp" with the following components:



```
src > JS App.js > default
1 // src/App.js
2 import React from 'react';
3 import ListofPlayers from './ListofPlayers';
4 import IndianPlayers from './IndianPlayers';
5
6 function App() {
7   // Toggle this flag variable between true and false to change the view output
8   const flag = true;
9
10  return (
11    <div style={{ maxWidth: '700px', margin: '40px auto', padding: '20px', border: '1px solid #ddd', borderRadius: '8px', box
12    <h1 style={{ textAlign: 'center', color: '#2c3e50', borderBottom: '2px solid #eee', paddingBottom: '15px' }}>
13      Cricket Analytics Dashboard
14    </h1>
15
16    <p style={{ background: '#f8f9fa', padding: '8px 12px', borderRadius: '4px', display: 'inline-block' }}>
17      Current View State: <strong>Flag = {flag.toString()}</strong>
18    </p>
19
20    /* Simple if-else conditional render block */
21    {flag ? <ListofPlayers /> : <IndianPlayers />}
22  </div>
23  );
24 }
25
26 export default App;
```



Create a React Application named “officespacerentalapp” which uses React JSX to create elements, attributes and renders DOM to display the page.

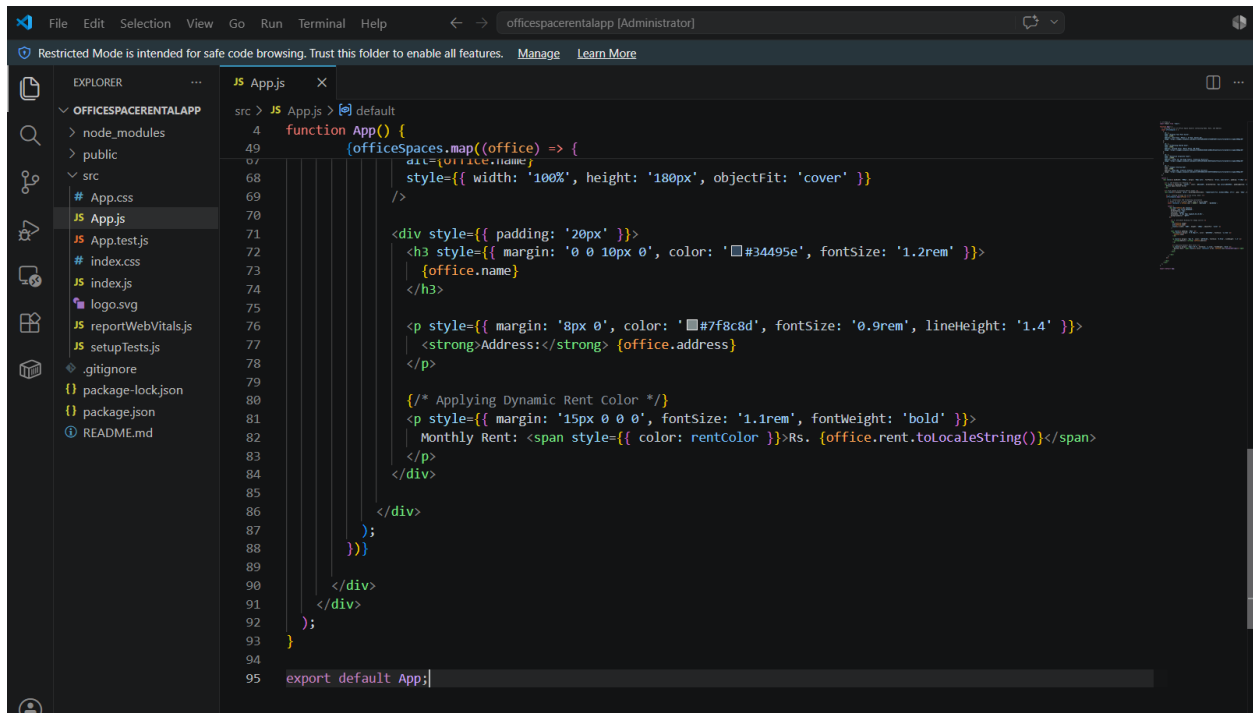
Create an element to display the heading of the page.

Attribute to display the image of the office space

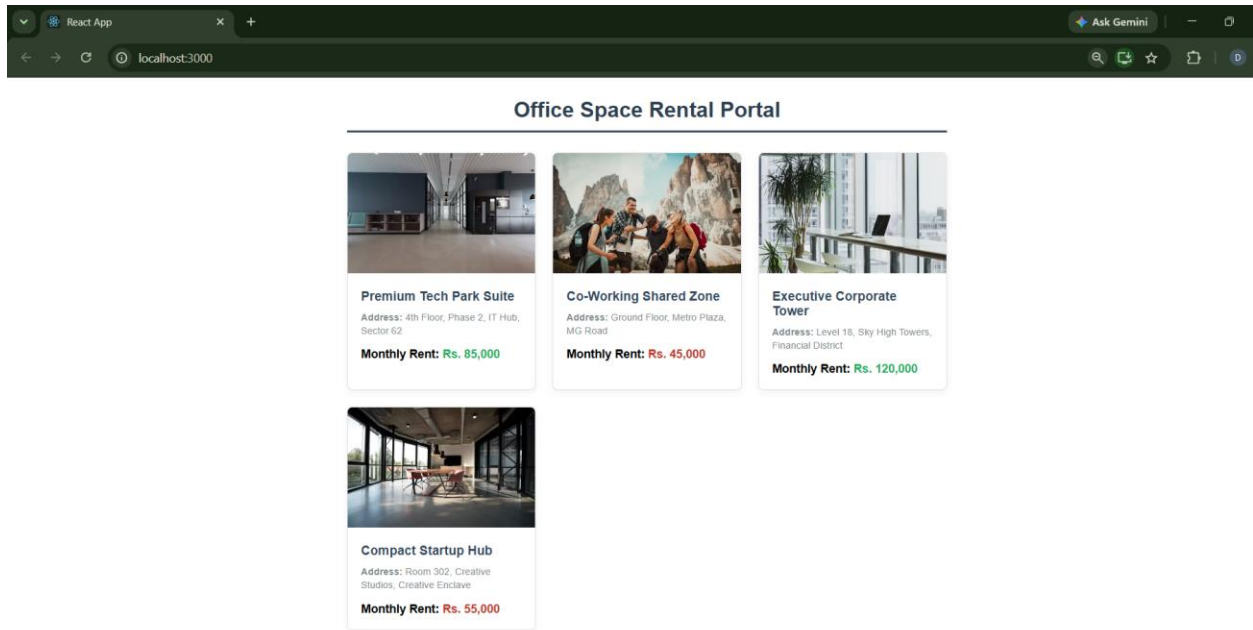
Create an object of office to display the details like Name, Rent and Address.

Create a list of Object and loop through the office space item to display more data.

To apply Css, Display the color of the Rent in Red if it's below 60000 and in Green if it's above 60000.



```
src > JS App.js > default
4  function App() {
49  (officeSpaces.map((office) => {
67      div={office.name;
68      style={{ width: '100%', height: '180px', objectFit: 'cover' }}
69  })
70  )
71
72  <div style={{ padding: '20px' }}>
73    <h3 style={{ margin: '0 0 10px 0', color: '#34495e', fontSize: '1.2rem' }}>
74      {office.name}
75    </h3>
76
77    <p style={{ margin: '8px 0', color: '#7f8c8d', fontSize: '0.9rem', lineHeight: '1.4' }}>
78      <strong>Address:</strong> {office.address}
79    </p>
80
81    <p style={{ margin: '15px 0 0 0', fontSize: '1.1rem', fontWeight: 'bold' }}>
82      Monthly Rent: <span style={{ color: rentColor }}>Rs. {office.rent.toLocaleString()}</span>
83    </p>
84  </div>
85  </div>
86  </div>
87  </div>
88  </div>
89  </div>
90  </div>
91  </div>
92  </div>
93  </div>
94  </div>
95  export default App;
```



ReactJS HOL – 10

Create a React Application “eventexamplesapp” to handle various events of the form elements in HTML.

File Edit Selection View Go Run Terminal Help eventexamplesapp [Administrator]

Restricted Mode is intended for safe code browsing. Trust this folder to enable all features. Manage Learn More

EXPLORER

EVENTEXAMPLESAPP

> node_modules

> public

> src

App.css

JS App.js

JS App.test.js

JS CurrencyConverter.js

index.css

JS index.js

logo.svg

JS reportWebVitals.js

JS setupTests.js

.gitignore

package-lock.json

package.json

README.md

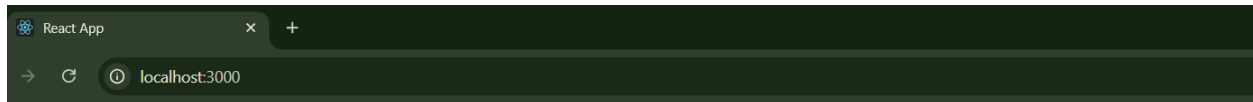
JS App.js

JS CurrencyConverter.js

src > JS App.js > default

```
1 // src/App.js
2 import React, { useState } from 'react';
3 import CurrencyConverter from './CurrencyConverter';
4
5 function App() {
6   const [counter, setCounter] = useState(0);
7
8   // 1. Methods for the Increment/Decrement Operations
9   const handleIncrement = () => {
10     setCounter(prev => prev + 1);
11   };
12
13   const sayHello = () => {
14     console.log("Hello! Counter increment event handled.");
15     alert("Hello! You successfully triggered multiple methods on a single click.");
16   };
17
18   // Wrapper method to invoke multiple operations on a single click event
19   const handleMultipleEvents = () => {
20     handleIncrement();
21     sayHello();
22   };
23
24   const handleDecrement = () => {
25     setCounter(prev => prev - 1);
26   };
27
28   // 2. Method that accepts a custom string argument
29   const sayWelcome = (message) => {
30     alert(`Message parameter received: "${message}"`);
31   };

```



React Event Showcase Portal

1. Counter and Multi-Method Handling

Current Counter Value: **1**

Increment & Say Hello Decrement

2. Passing Parameters to Event Handlers

Say Welcome

3. Synthetic Event (OnPress Simulation)

Trigger OnPress

Currency Converter (INR to EUR)

Enter Amount in Indian Rupees (Rs):

Convert

Equivalent Amount: € 100.00 Euros

ReactJS HOL – 11

Create a React Application named “ticketbookingapp” where the guest user can browse the page where the flight details are displayed whereas the logged in user only can book tickets. The Login and Logout buttons should accordingly display different pages. Once the user is logged in the User page should be displayed. When the user clicks on Logout, the Guest page should be displayed.

```
File Edit Selection View Go Run Terminal Help ticketbookingapp [Administrator]
Restricted Mode is intended for safe code browsing. Trust this folder to enable all features. Manage Learn More

EXPLORER
TICKETBOOKINGAPP
  > node_modules
  > public
  > src
    # App.css
    JS App.js
    JS App.test.js
    JS GuestPage.js
    # index.css
    JS index.js
    JS logo.svg
    JS reportWebVitals.js
    JS setupTests.js
    JS UserPage.js
    .gitignore
    package-lock.json
    package.json
    README.md
  > OUTLINE
  > TIMELINE

src > JS App.js > App
1 // src/App.js
2 import React, { useState } from 'react';
3 import GuestPage from './GuestPage';
4 import UserPage from './UserPage';
5
6 function App() {
7   // state variable to track if user is logged in
8   const [isLoggedIn, setIsLoggedIn] = useState(false);
9
10  return (
11    <div style={{ maxWidth: '800px', margin: '40px auto', padding: '25px', border: '1px solid #e2e8f0', borderRadius: '8px', b
12      <div style={{ display: 'flex', justify-content: 'space-between', align-items: 'center', borderBottom: '3px solid #2c3e50', padding-bottom: '10px' }}>
13        <h1 style={{ color: '#2c3e50', margin: 0, fontSize: '1.8rem', fontFamily: 'sans-serif' }}>
14          FlyHigh Ticket Booking Portal
15        </h1>
16        <div style={{ display: 'flex', gap: '10px' }}>
17          <button
18            onClick={() => setIsLoggedIn(false)}
19            style={{ padding: '10px 20px', backgroundColor: '#e74c3c', color: 'white', border: 'none', borderRadius: '4px', fo
20          > Logout
21        </button>
22        <button
23          onClick={() => setIsLoggedIn(true)}
24          style={{ padding: '10px 20px', backgroundColor: '#3498db', color: 'white', border: 'none', borderRadius: '4px', fo
25        > Login
26      </button>
27    </div>
28  </div>
29  </div>
30  </div>
31  </div>
32  </div>
33  </div>
34  </div>
35  </div>
36  </div>
```

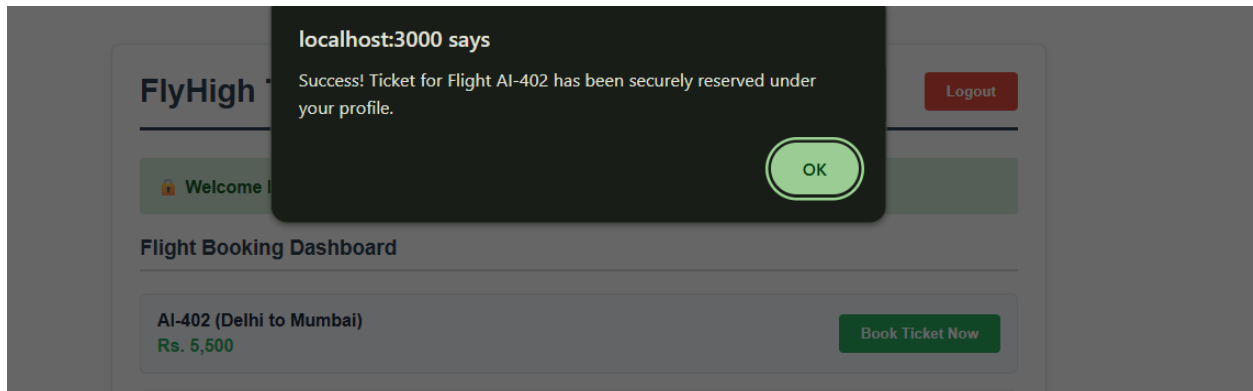
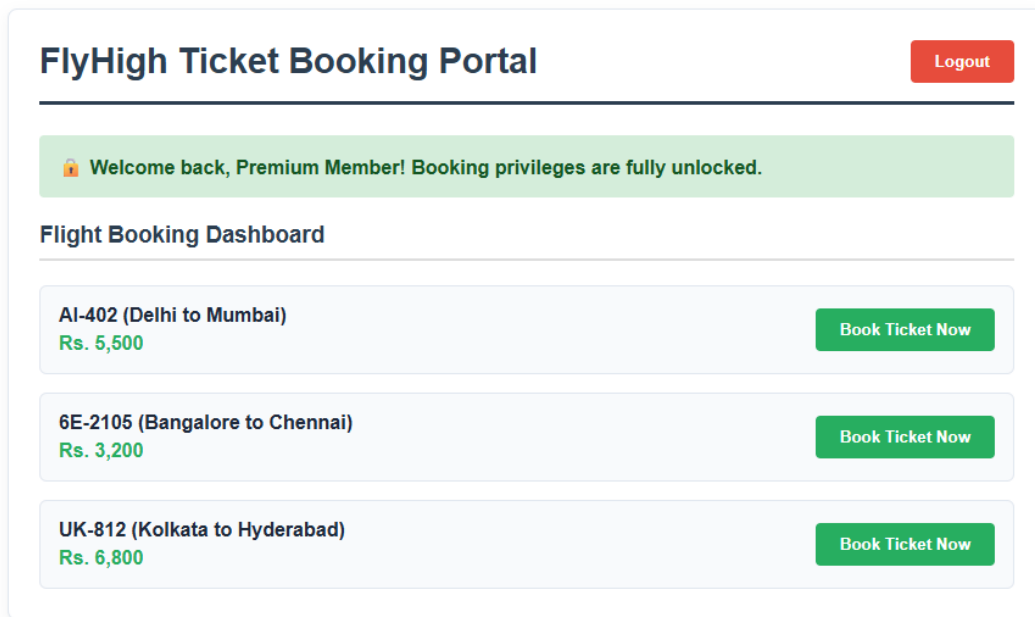
FlyHigh Ticket Booking Portal

Login

You are viewing as a Guest. Please log in to book your flight tickets.

Available Flights (Browse Mode)

Flight No	Route	Departure	Standard Fare
AI-402	Delhi (DEL) → Mumbai (BOM)	08:30 AM	Rs. 5,500
6E-2105	Bangalore (BLR) → Chennai (MAA)	11:15 AM	Rs. 3,200
UK-812	Kolkata (CCU) → Hyderabad (HYD)	04:45 PM	Rs. 6,800



ReactJS HOL - 12

Create a React App named "bloggerapp" in with 3 components.

1. Book Details
2. Blog Details
3. Course Details

Implement this with as many ways possible of Conditional Rendering.

File Edit Selection View Go Run Terminal Help bloggerapp [Administrator]

Restricted Mode is intended for safe code browsing. Trust this folder to enable all features. Manage Learn More

EXPLORER

BLOGGERAPP

> node_modules

> public

> src

App.css

JS App.js

JS App.test.js

JS BlogDetails.js

JS BookDetails.js

JS CourseDetails.js

index.css

JS index.js

logo.svg

reportWebVitals.js

JS setupTests.js

.gitignore

package-lock.json

package.json

README.md

src > JS App.js > default

1 // src/App.js

2 import React, { useState } from 'react';

3 import BookDetails from './BookDetails';

4 import BlogDetails from './BlogDetails';

5 import CourseDetails from './CourseDetails';

6

7 function App() {

8 // State for the main navigation view (Switch Case Demo)

9 const [currentView, setCurrentView] = useState('book');

10

11 // State variables for showing/hiding extra panels (Ternary and Short-Circuit Demo)

12 const [showQuickTips, setShowQuickTips] = useState(true);

13 const [isPremiumUser, setIsPremiumUser] = useState(false);

14

15 // --- APPROACH 1: Switch/Case Conditional Function Block ---

16 const renderMainComponent = () => {

17 switch (currentView) {

18 case 'book':

19 | return <BookDetails />;

20 case 'blog':

21 | return <BlogDetails />;

22 case 'course':

23 | return <CourseDetails />;

24 default:

25 | return <p>Select an option to view details.</p>;

26 }

27 };

Blogger Knowledge Hub

Book Info

Blog Info

Course Info

Method 1: Switch/Case Component Function Injection

Featured Book Details

Title: Mastering React Design Patterns

Author: Robin Wieruch

Pages: 340 pages

Method 2: Short-Circuit Evaluation (&&)

Hide Tips Panel

 **Quick Tip:** Use conditional rendering to completely mount/unmount parts of your application DOM dynamically.

Method 3: Inline Ternary Operator (? :)

☒ Toggle Premium Profile Unlocks

 **Pro Feature:** Full access to premium code repository downloads is unlocked!

Blogger Knowledge Hub

Book Info

Blog Info

Course Info

Method 1: Switch/Case Component Function Injection



Recommended Course Material

Course: Full-Stack Modern Web Engineering

Duration: 45 Hours

Platform: Cognizant Learning Academy

Method 2: Short-Circuit Evaluation (&&)

Show Tips Panel

Method 3: Inline Ternary Operator (? :)

☒ Toggle Premium Profile Unlocks

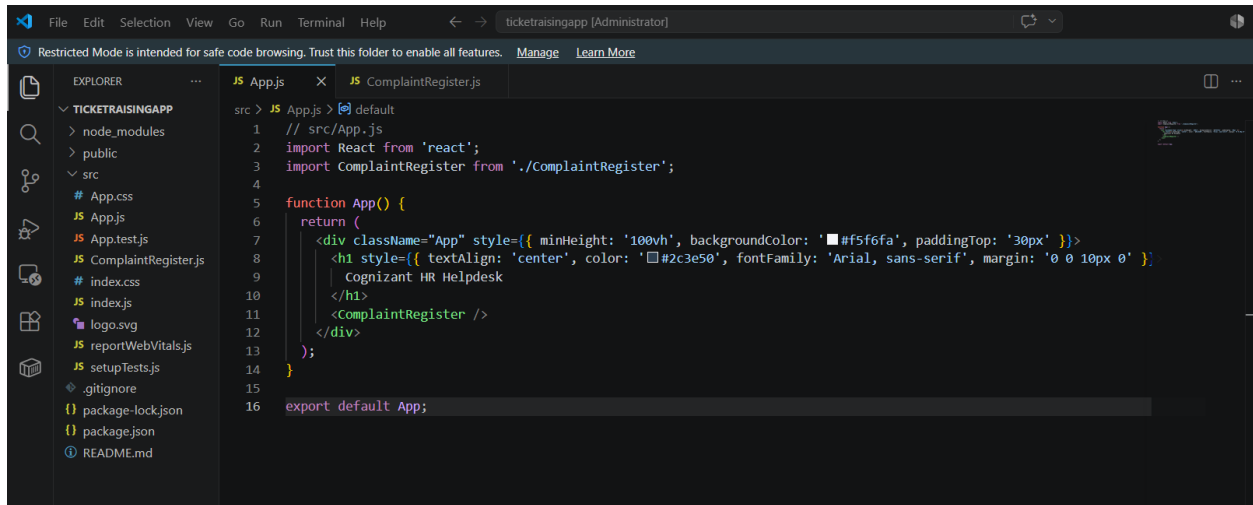


Pro Feature: Full access to premium code repository downloads is unlocked!

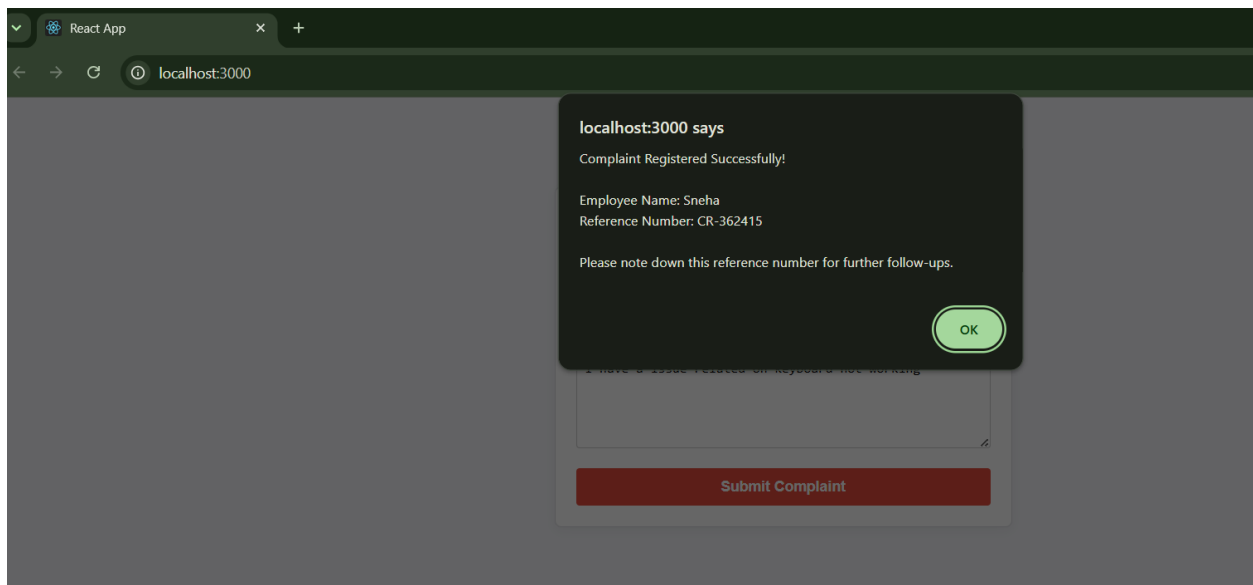
ReactJS HOL - 13

Create a React App named “ticketraisingapp” which will help to raise a complaint and get it resolved.

Create a component named “ComplaintRegister” with a form containing a textbox to enter the employee name and a textarea to enter the complaint. Use “handleSubmit” event of the button to submit the complaint and generate a Reference number for further follow ups in the alert box.



```
1 // src/App.js
2 import React from 'react';
3 import ComplaintRegister from './ComplaintRegister';
4
5 function App() {
6   return (
7     <div className="App" style={{ minHeight: '100vh', backgroundColor: '#f5f6fa', paddingTop: '30px' }}>
8       <h1 style={{ textAlign: 'center', color: '#2c3e50', fontFamily: 'Arial, sans-serif', margin: '0 0 10px 0' }}>
9         Cognizant HR Helpdesk
7       </h1>
10      <ComplaintRegister />
11    </div>
12  );
13 }
14
15 export default App;
```

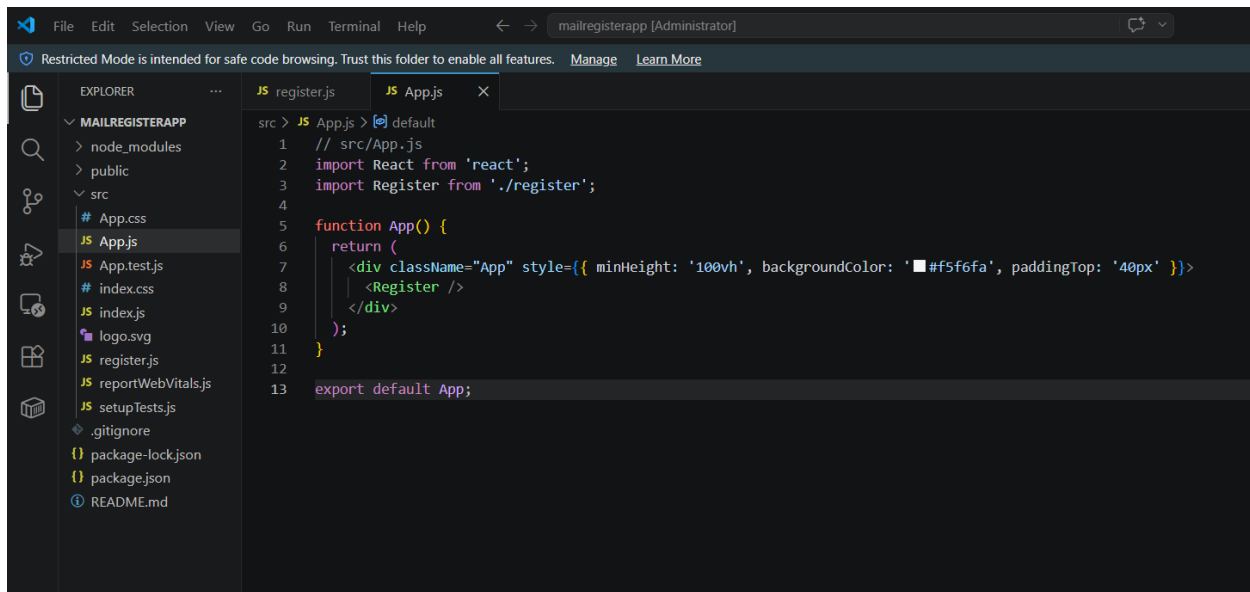


ReactJS HOL – 14

Create a React App named “mailregisterapp” which will have a component named “register.js”. Create a form which accepts the name, email and password and validate the fields as per the following:

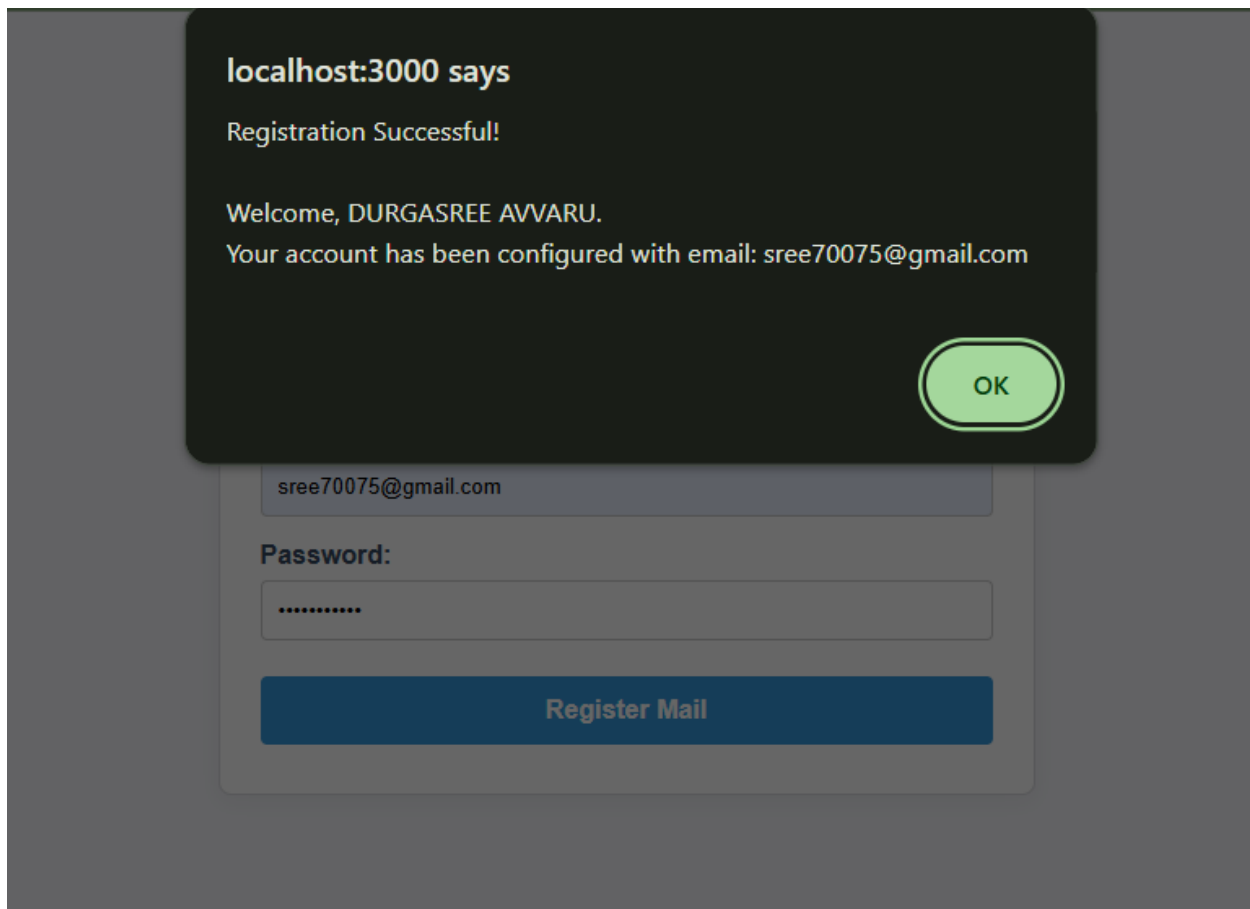
1. Name should have atleast 5 characters
2. Email should have @ and .
3. Password should have atleast 8 characters.

Ensure that validations are implemented through eventhandle and eventsubmit of a form.



The screenshot shows the Visual Studio Code editor interface. The Explorer panel on the left displays the file structure of a project named 'MAILREGISTERAPP'. The file 'App.js' is selected. The main editor area shows the code for 'App.js', which imports 'React' and 'Register' from './register', defines an 'App' function that returns a JSX element, and exports it as the default.

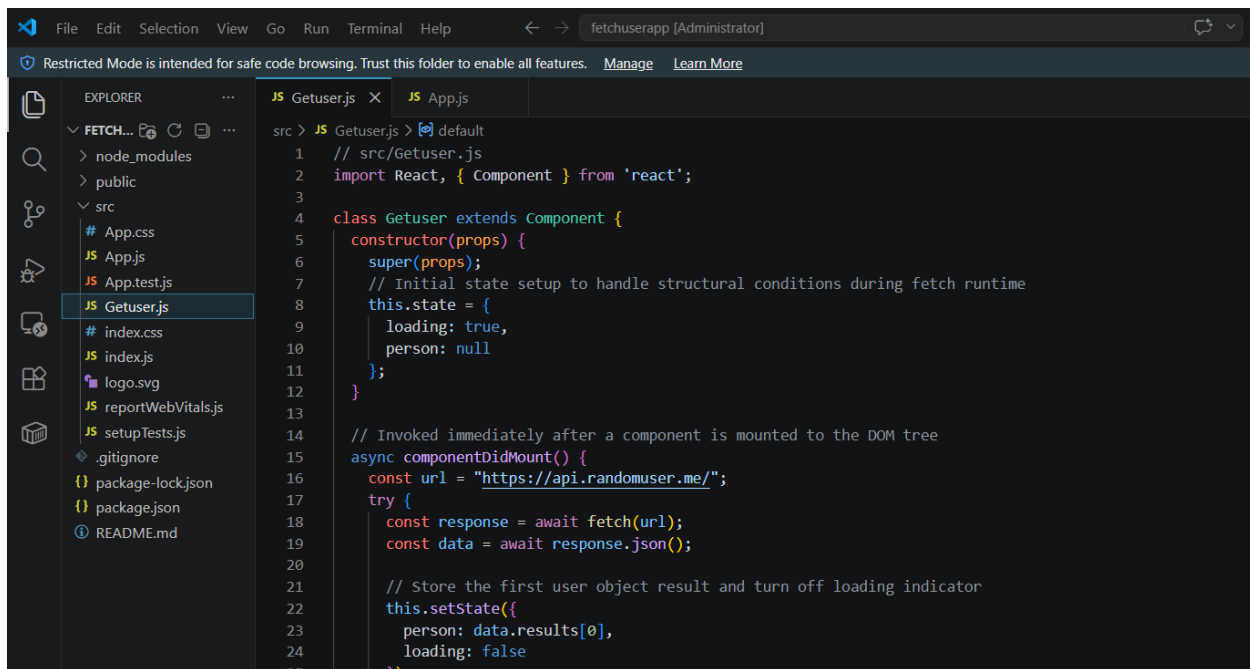
```
src > JS App.js > default
1 // src/App.js
2 import React from 'react';
3 import Register from './register';
4
5 function App() {
6   return (
7     <div className="App" style={{ minHeight: '100vh', backgroundColor: '#f5f6fa', paddingTop: '40px' }}>
8       <Register />
9     </div>
10  );
11 }
12
13 export default App;
```



ReactJS HOL – 15

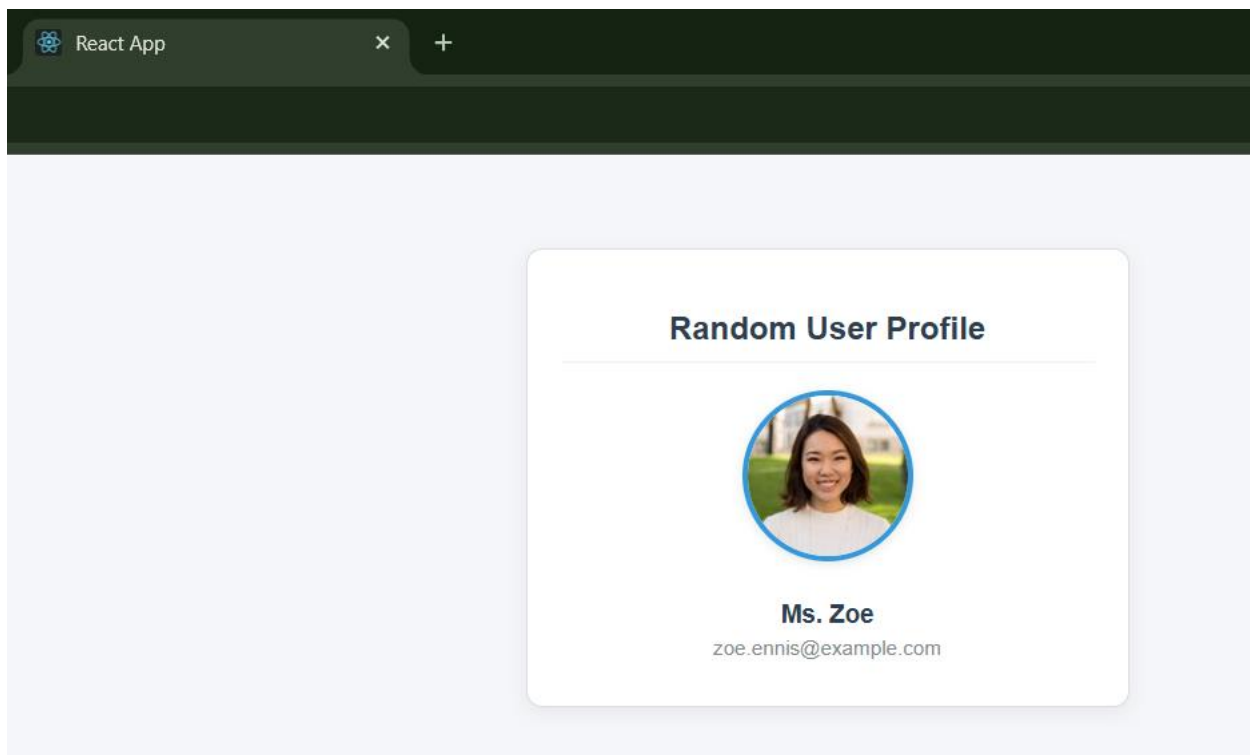
Create a React Application “fetchuserapp” which will retrieve the user details from <https://api.randomuser.me/> and display the title, firstname and image of a user.

Create a component named “Getuser” and in the asynchronous method “ComponentDidMount ()” invoke the URL using fetch method and the response can be displayed in the render method of the component.



The screenshot shows a VS Code editor window with the file explorer on the left and the code editor on the right. The file explorer shows a project structure with a 'src' folder containing 'App.js', 'App.test.js', 'Getuser.js', 'index.css', 'index.js', 'logo.svg', 'reportWebVitals.js', 'setupTests.js', '.gitignore', 'package-lock.json', 'package.json', and 'README.md'. The code editor shows the 'Getuser.js' file with the following code:

```
1 // src/Getuser.js
2 import React, { Component } from 'react';
3
4 class Getuser extends Component {
5   constructor(props) {
6     super(props);
7     // Initial state setup to handle structural conditions during fetch runtime
8     this.state = {
9       loading: true,
10      person: null
11    };
12  }
13
14  // Invoked immediately after a component is mounted to the DOM tree
15  async componentDidMount() {
16    const url = "https://api.randomuser.me/";
17    try {
18      const response = await fetch(url);
19      const data = await response.json();
20
21      // Store the first user object result and turn off loading indicator
22      this.setState({
23        person: data.results[0],
24        loading: false
25      });
26    } catch (error) {
27      console.log(error);
28    }
29  }
30
31  render() {
32    return (
33      <div>
34        <h2>Random User Profile</h2>
35        <img alt="User Profile Picture" data-bbox="566 658 671 741" />
36        <p>Ms. Zoe</p>
37        <p>zoe.ennis@example.com</p>
38      </div>
39    );
40  }
41}
```



As an intern at OpenAI you are assigned the task of creating and testing a React application which will fetch and display a list of repository names for a given user.

Create a new React application using *create-react-app* tool and name it as “gitclientapp”.

```
src > JS App.js > default
1 // src/App.js
2 import React, { useState } from 'react';
3 import GitClient from './GitClient';
4
5 function App() {
6   const [username, setUsername] = useState('techiesyed');
7   const [repos, setRepos] = useState([]);
8   const [loading, setLoading] = useState(false);
9   const [error, setError] = useState('');
10
11   const fetchRepos = async () => {
12     if (!username.trim()) return;
13     setLoading(true);
14     setError('');
15     try {
16       const repoNames = await GitClient.getRepositories(username);
17       setRepos(repoNames);
18     } catch (err) {
19       setError('Could not retrieve repositories. Verify the username.');
```

GitHub Repo Explorer

techiesyed

Fetch Repos

Repositories for techiesyed:

- ArrayListDemo
- ArrayListDemo01
- AzureDevopsDemoProductsApi
- CICDMvcDemo
- CleanArchitecture
- ContosoUniversity
- DelagatesDemo
- EventsDemo
- FISGlobal-0721
- GenericCollections
- GenericsDemo
- GitHubDemoApp
- GitTestFolder
- HashTableDemo
- HelloJenkins
- HelloWorldJenkinsAngular
- Layered-Architecture-Demo
- musical-octo-carnival
- mvc03
- Mvvm-With-WPF
- MyApi
- ReflectionDemo
- stunning-broccoli
- techiesyed
- TestRepo
- Webinar-Automation-testing-Selenium-CSharp
- WhyUseGenerics